Build guide 1: tools and steel

DIY CNC epoxy-granite mill · October 2026

0. Tools, space and safety

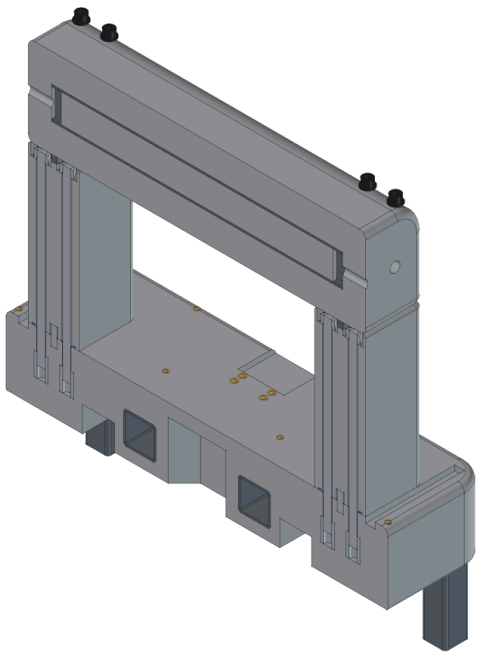
- **Space:** a level bench or floor area of about 1.2 x 1.5 m at 20-25 °C for a week. Epoxy cures slowly and badly below 18 °C.
- **Tools:** drill press, chop saw or hacksaw, angle grinder with flap disc, taps (M4, M5, M6), drill/mixer with a paddle (or a small cement mixer), kitchen scale + a 50 kg scale, vibration source (a table vibrator, a concrete poker, or a reciprocating sander with the pad removed pressed against the mold), straightedge (at least 600 mm, ideally ground), feeler gauges, dial indicator with a magnetic base, precision level, torque wrench.
- **Safety:** nitrile gloves and long sleeves for epoxy (skin sensitisation is permanent); an N95 dust mask when pouring the whiting; a P100 respirator when grinding or sanding cured EG (the quartz sand and gravel in it make silica dust; work wet if you can); eye protection when cutting steel. Mix in a ventilated space.
- **3D printing (PETG):** the mold kit (brackets, saddles, holders, bushings), the core posts, formers and fillers, the parts that get cast in (spigot sleeves, tube plugs, drain liner) and the machine's printed parts. All of it is in `print/`, split to fit a Bambu A1 mini, and listed in the print guide: 1.3 kg of PETG (sliced), inside two 1 kg rolls, and about 110 hours of printing. The base pocket and socket cores are cut from one sheet of 2 in XPS foam board instead. Print one bracket and one saddle first and check them on your board; print the sleeves early and test-fit them on your actual tube.
- **Foam cutting:** a fine-tooth saw or a hot-wire cutter, a long knife, a sanding block, packing tape and double-sided carpet tape.

1. Buy and prepare the steel

Nothing you order needs cutting: buy the tube at these lengths, and order the risers, the top plates and the aluminum plates cut to size. Only the 1/2-13 threaded rod is cut.

1. **Square tube** (plain steel, not galvanized; 0.120 or 1/8 in wall; remnants are fine):
   - 2 x 2 in: 4 x 12 in (base; they also take the split spigots), 2 x 18 in (beam).
   - 1-1/2 x 1-1/2 in: 4 x 12 in (upright posts), 6 x 6 in (2 base spigots + 4 legs).
2. **Check the steel when it arrives.** A supplier's saw cut is good to about 1/8 in, and square tube is about 0.5 mm off its nominal size, with a weld seam down the inside. None of that needs fixing, because each piece is set from one end and the other end has room (the "Steel cut to length" table in `docs/5-design/report.txt` lists it):
   - **Length:** anything within 1/8 in of the listed length is fine. Mark one end of each 2 in base tube as its split end, and mark the middle of each spigot.
   - **Ends:** square within a couple of degrees (a 2 in end no more than 1.5 mm out across it). Knock off burrs.
   - **Size:** caliper the outside of each size of tube, and the inside of a 2 in tube away from the seam. The printed parts that fit on or into tube (bushings, leg holders, sleeves, plugs, locators) have 0.5 mm a side and grooves for the seam, so they fit tube that's up to 0.5 mm off. If yours is further off, enter the sizes in `frame_tube` / `post_tube` in `model/build_cnc.py` (with `tube_fit` at 0.25 for a closer fit), re-run it and print those parts from the new files.
   - **Test fit:** a printed sleeve on a spigot, both into a 2 in base tube. If it's tight, file the seam. If it's loose, wrap the sleeve in a turn of tape.

3. **Drill the beam tubes:** each 18 in tube gets 3/4 in holes straight through its top and bottom walls where the inner beam cap screws pass, 4 holes per tube. Measure the tube, mark its middle, and mark the holes 203 mm either side of that mark on the cap-screw line (7 mm off the tube's center line, toward the outside of the beam). Measuring from the middle splits any length error between the two ends, and the 3/4 in holes give the 1/2 in rods that form the bolt holes 3 mm of play a side. Use a drill press, a step drill or a hole saw.
4. **Top plates** (3/8 in steel, 2 pieces 68 x 168 mm, ordered cut to size): drill and tap the four 3/8-16 beam-screw holes and a 14 mm dowel hole in each (oversize: the beam's dowel is grouted into it at assembly), and cut the tie-rod pockets (see the section picture below). Do this before casting.
5. **X support pads** (1/4 x 2-1/2 in flat bar, 2 pieces, about 49 and 36 mm long): drill and tap M6 at the BK10 and BF10 bolt positions (46 mm apart across the axis; along it, 13 mm apart for the BK10). These end up flush in the beam face and carry the X ballscrew bearings.
6. **Rail risers** (cold-rolled 1018, 1-1/4 in wide, ordered cut to the rail lengths): Y 3/4 in x 600 mm (2), X 5/8 in x 400 mm (2), Z 5/8 in x 300 mm (2). The rails are used at full length.
    - Clamp each rail on its riser, centerd, and use the rail's own holes as the drill template; drill and tap M4 (60 mm pitch). Mark which rail goes on which riser.
    - Y and X risers also get the hold-down holes, midway between rail holes and under the rail: 5.5 mm through with a 9.5 mm x 4.5 mm deep counterbore from the top. Positions from the riser center: Y at 0, ±120 and ±240 mm, X at ±30 and ±150 mm.
    - Stone or lightly sand off burrs; keep the top faces flat.
7. **Clean all cast-in steel:** wire-brush or flap-disc to dull gray metal, then degrease with acetone just before casting. If it will sit more than a day or two, give it a thin coat of epoxy primer. Paint only the part of each leg that will hang below the base (after casting).
8. **Threaded rod and hardware:** cut the 1/2-13 rods (per upright 2 x 293, 1 x 213 and 1 x 102 mm; 2 x 358 mm for the base draw rods), deburr the ends. Have the inserts ready: 18 x M5 (12-16 mm long) and 6 x M6 x 15 mm cast-in inserts, plus 11 x M5 for the enclosure.



The cast-in steel, inserts and printed liners for each base half (castings drawn see-through):

